Booklet

1

CHEMISTRY 3AB

MULTIPLE CHOICE QUESTIONS BOOKLET

Time allowed for this paper
Reading time before commencing work: ten (10) minutes
Working time for paper: three (3) hours

Materials required/recommended for this paper
To be provided by the supervisor
Booklet 1 – Multiple choice questions booklet
Booklet 2 - Question/Answer booklet with multiple choice answer sheet.
Chemistry Data sheet

To be provided by the candidate
Standard items: pens, pencils, eraser, correction fluid/tape, ruler, highlighters
Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course

Important note to candidates
No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks Available	Percentage of exam
Section One: Multiple-choice	25	25	30	**50**	25
Section Two: Short answer	10	10	70	**70**	35
Section Three: Extended answer	4	4	80	**80**	40
			Total	**200**	100

Instructions to candidates

1. Write your name and your teacher's name on **Booklet 2 only**.

2. The rules for the conduct of Western Australian external examinations are detailed in the *Student Information Handbook 2013*. Sitting this examination implies that you agree to abide by these rules.

3. Answer the questions according to the following instructions.

   Section One: Answer all questions on the Multiple-choice answer sheet attached to the front of Booklet 2. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

   Sections Two and Three: Write your answers in Booklet 2.

4. When calculating numerical answers, show your working or reasoning clearly unless instructed otherwise.

5. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

6. Spare pages are included at the end of Booklet 2. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages in planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
     Fill in the number of question(s) that you are continuing to answer at the top of the page.

Section One: Multiple-choice **25% (50 marks)**

This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 30 minutes.

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1. What is the number of protons, electrons and neutrons in the $^{81}_{35}Br^{5+}$ ion?

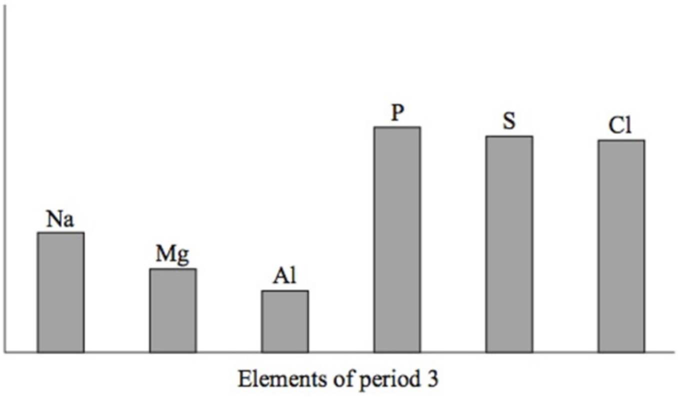
(a) 35 protons 40 electrons 46 neutrons

(b) 35 protons 30 electrons 46 neutrons

(c) 35 protons 30 electrons 81 neutrons

(d) 35 protons 40 electrons 81 neutrons

2. The horizontal axis of the bar chart below represents the elements of period 3 from sodium to chlorine (excluding silicon). What could the vertical axis represent?



(a) Melting point of the element

(b) Electronegativity of the bonded atom

(c) Ionic radius of the most common ion

(d) First ionization energy in the gaseous state

3. Which of the following covalent bonds would be expected to have the **greatest** polarity?

(a) N – F

(b) P – F

(c) S – F

(d) O – F

Questions 4 and **5** are based upon the table below, which gives the first eight ionisation energies (in kJ mol^{-1}) of four elements, all of which are in the first 20 elements of the Periodic Table.

Element	**1st**	**2nd**	**3rd**	**4th**	**5th**	**6th**	**7th**	**8th**	
(i)	590	1,145	4,912	6,491	8,153	10,496	12,270	14,206	2
(ii)	577	1,820	2,740	11,600	14,800	18,400	22,400	27,500	3
(iii)	495	4,562	6,910	9,543	13,354	16613	20,117	25,496	1
(iv)	1000	2260	3390	4560	6990	8490	27,200	37,100	6

4. Which element would you expect to have the highest melting point?

(a) (i)

(b) (ii)

(c) (iii)

(d) (iv)

5. Which element would you expect to have the lowest electronegativity?

(a) (i)

(b) (ii)

(c) (iii)

(d) (iv)

6. Which of the following structural features is common to both diamond and graphite?

(a) Covalent bonds between carbon atoms

(b) Delocalised electrons

(c) Dipole-dipole interaction

(d) Covalent dipole attraction

7. Which of the following properties do **not** have an increasing trend across Period 3 of the Periodic Table?

(i) Atomic number
(ii) Atomic radius
(iii) Electronegativity
(iv) Ionization energy
(v) Number of electron shells

(a) (i) and (iv)

(b) (i), (iii) and (iv)

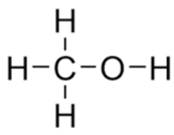
(c) (ii) and (v)

(d) (i), (iii), (iv) and (v)

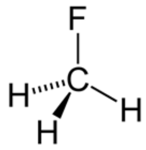
See next page

8. In which of the following liquids would you expect hydrogen bonding to be present between their molecules

Hydrogen fluoride



Methanol



Fluoromethane

(a) all of them.

(b) hydrogen fluoride and methanol only.

(c) hydrogen fluoride only

(d) methanol only

9. Which of the following **best** explains why water has a lower boiling point than sodium chloride?

(a) The covalent bonds in water are weaker than the ionic bonds in sodium chloride.

(b) The intermolecular forces between water molecules are weaker than ionic bonds in sodium chloride.

(c) The molecular mass of water (18.016) is less than the molecular mass of sodium chloride (58.44).

(d) One mole of water occupies a smaller volume (18.02 mL) than one mole of sodium chloride at the same temperature and atmospheric pressure (27.01 mL).

10. The reaction between sulfur dioxide gas and oxygen gas to form sulfur trioxide gas is part of the synthesis of sulfuric acid by the Contact Process. It is catalysed by vanadium (V) oxide and after a certain time, equilibrium in the closed system is reached. Consider the following statements about the role of the catalyst in this reaction.

(i) The catalyst allows the reaction to reach equilibrium sooner by increasing the rate of both the forward and reverse reactions.

(ii) The catalyst lowers the activation energy barrier for both the forward and reverse reactions.

(iii) The catalyst increases the mass of products compared to the mass of reactants when equilibrium is reached.

Which of the following statements is correct?

(a) Only statement (i) above is true.

(b) All the above statements are true.

(c) None of the above statements is true.

(d) Only statements (i) and (ii) above are true.

11. Consider the following reaction at equilibrium:

$$2SO_{2(g)} + O_{2(g)} \leftrightarrow 2SO_{3(g)} \qquad \Delta H = -196 \text{ kJ mol}^{-1}$$

Which of the following would increase the rate of the **reverse** reaction, once equilibrium is re-established?

(i) Adding more SO_2

(ii) Adding more SO_3

(iii) Increasing the temperature

(iv) Decreasing the volume of the container

(a) (i), (ii) and (iv)

(b) (i), (iii) and (iv)

(c) (ii), (iii) and (iv)

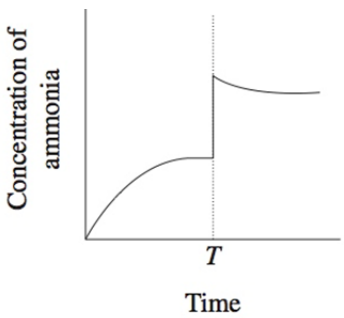
(d) All of them

12. The following equation shows an equilibrium established in the synthesis of ammonia from its component gases:

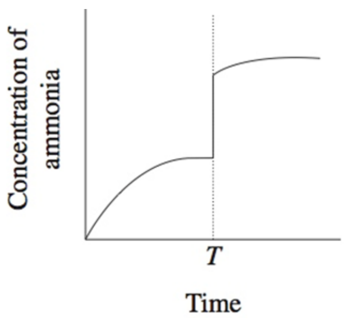
$$N_{2(g)} + 3H_{2(g)} \leftrightarrow 2NH_{3(g)}$$

If the volume of the reaction chamber is suddenly halved (i.e. a reduction of 50%) at time ***T*,** which of the following graphs best depicts changes in the concentration of **ammonia** over time?

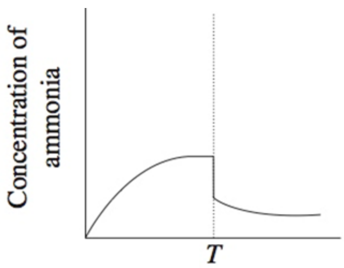
(a)



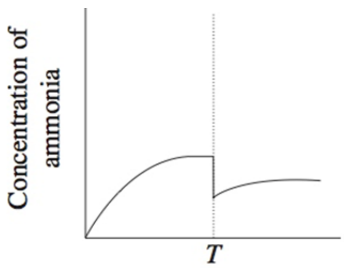
(b)



(c)



(d)



Questions 13, 14 and 15 refer to the following reaction between phosphorus trichloride and chlorine to form phosphorus pentachloride.

$$PCl_{3(g)} + Cl_{2(g)} \leftrightarrow PCl_{5(g)} \qquad \Delta H = -124 \text{ kJ mol}^{-1}$$

13. If phosphorus trichloride and chlorine were placed in a sealed and insulated container together with a catalyst, which of the following would **not** cause an increase in the rate at which equilibrium would be obtained?

(a) Increasing the volume of the container

(b) Increasing the state of subdivision of the catalyst

(c) Increasing the temperature

(d) All of the above

14. Which of the following statements is/are **true** when the system is at equilibrium?

i. Reactants are no longer turning into products

ii. The concentration of chlorine in the container stays unchanged.

iii. Adding a catalyst would not affect the proportions of the reaction mixture

(a) (iii) only

(b) (i) and (ii) only

(c) (ii) and (iii) only

(d) (i), (ii) and (iii)

15. Which of the following changes would **not** be observed once equilibrium is re-established if a little phosphorus trichloride were added at constant temperature to the equilibrium mixture above?

(a) The rate of the forward reaction would increase.

(b) The rate of the reverse reaction would increase.

(c) The concentration of phosphorus trichloride would increase.

(d) The value of the equilibrium constant (K) would decrease.

16. When chloride ions are added to a solution containing $Co(H_2O)_6^{2+}$ the following equilibrium is established:

$$Co(H_2O)_6^{2+}{}_{(aq)} + 4Cl^-{}_{(aq)} \leftrightarrow CoCl_4^{2-}{}_{(aq)} + 6H_2O_{(l)}$$

Pink ***Blue***

Solutions containing $Co(H_2O)_6^{2+}$ and Cl^- are frequently violet in colour owing to the presence of significant amounts of both $Co(H_2O)_6^{2+}$ and $CoCl_4^{2-}$.

Which of the following statements concerning such solutions is **true**?

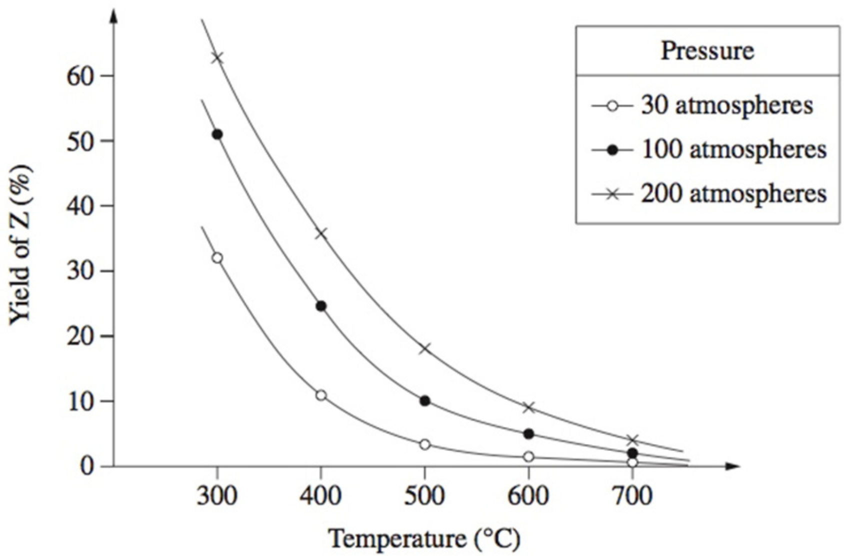
(a) If the forward reaction is endothermic, cooling the solution will make the colour turn pink.

(b) Adding a large amount of solid sodium chloride to the solution will make the colour turn pink.

(c) Diluting the solution with water will make the colour turn blue.

(d) If the forward reaction is exothermic, heating the solution will make the colour turn blue.

17. The graph below represents the yield of an equilibrium reaction at different temperature and pressure conditions inside a reaction vessel.



forward exothermic & more gas particles

Which of the following reactions could produce the trends shown in the graph?

(a) $X_{(g)} + Y_{(g)} \leftrightarrow 3Z_{(g)}$ $\Delta H = +250kJ$

(b) $X_{(g)} + Y_{(g)} \leftrightarrow 2Z_{(g)}$ $\Delta H = -250kJ$

(c) $2X_{(g)} + 2Y_{(g)} \leftrightarrow Z_{(g)}$ $\Delta H = +250kJ$

(d) $4X_{(g)} + 2Y_{(g)} \leftrightarrow 3Z_{(g)}$ $\Delta H = -250kJ$

18. Consider a solution of lead (II) chloride at 25°C in equilibrium with some solid lead (II) chloride at the bottom of the solution.

$$Pb^{2+}_{(aq)} + 2Cl^{-}_{(aq)} \leftrightarrow PbCl_{2(s)}$$

If the forward reaction is exothermic, which of the following is the **best** way to increase the concentration of the lead and chloride ions in the solution?

(a) Allowing some of the water to evaporate and then leaving the system to reach equilibrium again.

(b) Heating the system to 30°C.

(c) Adding some more solid lead (II) chloride.

(d) Sealing the container and vigorously shaking the solution.

19. Which one the following solutions has the highest concentration of dissolved ions at 25°C?

(a) 0.4 $molL^{-1}$ CH_3COOH

(b) 0.3 $molL^{-1}$ $Mg(NO_3)_2$

(c) 0.5 $molL^{-1}$ $Al(OH)_3$

(d) 0.4$molL^{-1}$ Na_3PO_4

20. Four students were asked to test a solution for the presence of a cation by using various anions. The students obtained these results:

Student	***Chloride***	***Sulfate***	***Carbonate***
Acka-Dacka	No precipitate	No precipitate	Precipitate
Beeba	Precipitate	Precipitate	No precipitate
Cindy-Ella	Precipitate	Precipitate	Precipitate
Darth	No precipitate	Precipitate	No precipitate

Each student concluded that Pb^{2+} was present.

Which student had results consistent with this conclusion?

(a) Acka-Dacka

(b) Beeba

(c) Cindy-Ella

(d) Darth

21. The following sets of solutions are mixed. In which of these will there be no visible reaction?

(a) Set 1 consisting of barium hydroxide, dilute sulfuric acid, potassium nitrate. ($BaSO_4$)

(b) Set 2 consisting of potassium phosphate, copper(II) sulfate and sodium chloride. ($Cu_3(PO_4)_2$)

(c) Set 3 consisting of sodium chloride, copper(II) sulfate and silver nitrate. (AgCl)

(d) Set 4 consisting of sodium iodide, barium nitrate and potassium hydroxide

22. The solubility of oxygen in seawater at 20°C is 14.4 mgL^{-1}. Which of the following is the concentration of oxygen in seawater in molL^{-1}?

$$14{\cdot}4\,\text{mg} = 0{\cdot}0144\,g$$
$$\frac{0{\cdot}0144}{32} = 0{\cdot}00045 = 4{\cdot}5 \times 10^{-4}$$

(a) 4.5×10^{-1} mol L^{-1}

(b) 9.0×10^{-1} mol L^{-1}

(c) 4.5×10^{-4} mol L^{-1}

(d) 9.0×10^{-4} mol L^{-1}

23. Which of the following correctly shows an acid- conjugate base pair?

$$\mathbf{HCO_3^-{}_{(aq)} + H_2O_{(l)} \leftrightarrow CO_3^{2-}{}_{(aq)} + H_3O^+{}_{(aq)}}$$

(a) HCO_3^- and H_3O^+

(b) H_2O and CO_3^{2-}

(c) HCO_3^- and CO_3^{2-}

(d) All of the above

24. Which one of the following species is acting as a **Lowry-Bronsted acid** when ammonia gas is dissolved in a salt solution containing water and fully dissolved sodium chloride.

$$NH_3(g) + NaCl(aq) + H_2O(l) \longrightarrow NH_4^+ + OH^- + NaCl(aq)$$

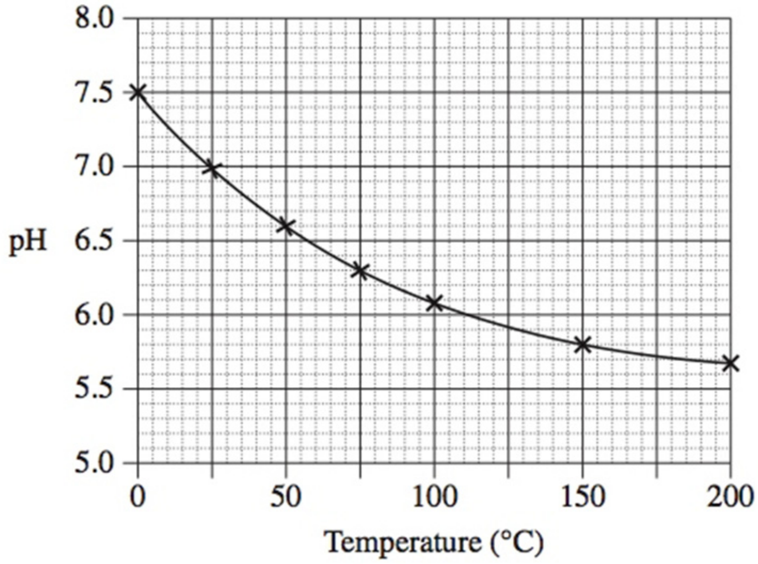
(a) NH_3

(b) H_2O

(c) Na^+

(d) Cl^-

25. The graph below shows the pH of a solution of a weak acid, HA, as a function of temperature.



What happens as the temperature decreases?

(a) HA becomes less ionised and the $H^{+}_{(aq)}$ concentration increases.

(b) HA becomes less ionised and the $H^{+}_{(aq)}$ concentration decreases.

(c) HA becomes more ionised and the $H^{+}_{(aq)}$ concentration increases.

(d) HA becomes more ionised and the $H^{+}_{(aq)}$ concentration decreases.

END OF BOOKLET 1

Kira College

CHEMISTRY 12 SEMESTER 1 EXAMINATION 2015

Booklet 2

Name:______________________ **Teacher:** _________________________

For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. For example, if b is your answer: ☐a ■b ☐c ☐d

If you make a mistake, place a cross through that square, do not erase or use correction fluid.
Shade your new answer.

For example, if b is a mistake and d is your correct answer: ☐a ☒b ☐c ■d

In the event that you then change your mind back to your original answer, you then cross out the second selection and then circle the first choice.

For example, if b was the first choice and d your second, but you changed your mind back and b is your answer: ☐a ☒b ☐c ☒d

Multiple Choice: Mark the answer by shading in the box to the LEFT of the letter you chose.

1. ☐a ■b ☐c ☐d

2. ☐a ☐b ■c ☐d

3. ☐a ■b ☐c ☐d

4. ☐a ☐b ■c ☐d

5. ☐a ☐b ■c ☐d

6. ■a ☐b ☐c ☐d

7. ☐a ☐b ■c ☐d

8. ■a ☐b ☐c ☐d

9. ☐a ■b ☐c ☐d

10. ☐a ☐b ☐c ■d

11. ☐a ☐b ☐c ■d

12. ☐a ■b ☐c ☐d

13. ■a ☐b ☐c ☐d

14. ☐a ☐b ■c ☐d

15. ☐a ☐b ☐c ■d

16. ■a ☐b ☐c ☐d

17. ☐a ☐b ☐c ■d

18. ☐a ■b ☐c ☐d

19. ☐a ☐b ☐c ■d

20. ☐a ☐b ■c ☐d

21. ☐a ☐b ☐c ■d

22. ☐a ☐b ■c ☐d

23. ☐a ☐b ■c ☐d

24. ☐a ■b ☐c ☐d

25. ☐a ■b ☐c ☐d